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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester M.Tech(CSE)(CBCS) Degree Examinations

December 2024 – January 2025

23CSE102 – ADVANCED COMPUTER NETWORKS

Duration: 3 Hours

Max. Marks:100

Note: Answer Five full questions choosing One full question from each Unit

		Marks	BT*	CO*	PO*
Unit – I					
1.	a) Demonstrate different topologies of network.	10	L*3	1	2
	b) Demonstrate TCP/IP Protocol suite.	10	L3	1	2
2.	a) Explain different layers of addressing in TCP/IP.	10	L2	1	1
	b) Demonstrate the different transmission strategies from IPv4 to IPv6.	10	L3	1	1
Unit – II					
3.	a) Develop process to process delivery and illustrate socket address.	10	L3	2	2
	b) Demonstrate connection establishment using three-way handshaking.	10	L2	2	2
4.	a) Construct TCP Segment format and explain.	10	L3	2	2
	b) Explain congestion and illustrate different categories of congestion control.	10	L2	2	1
Unit – III					
5.	a) Explain Session Initiation Protocol.	10	L2	3	2
	b) Illustrate three network-level approaches to supporting multimedia Applications.	10	L2	3	2
6.	a) Interpret forward error correction (FEC) and interleaving.	10	L2	3	1
	b) Explain HTTP streaming and illustrate the interaction between client and server for HTTP streaming.	10	L2	3	1
Unit – IV					
7.	a) Demonstrate 3G cellular network architecture.	10	L3	4	2
	b) Explain indirect routing to mobile node.	10	L2	4	1
8.	a) Explain Code division multiple access with example.	10	L2	4	2
	b) Demonstrate IEEE 802.11 Frame.	10	L3	4	2
Unit – V					
9.	a) Explain the importance of SDN.	10	L3	5	2
	b) Illustrate the use cases for input traffic monitoring.	10	L2	5	1
10.	a) Demonstrate wireless sensor network application in industrial automation.	10	L3	5	2
	b) Explain hardware and software components of sensor nodes.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
